Vector XSW

Step-by-step derivation

1 Equations and ansatz

We use **Gaussian units**: no ε_0, μ_0, or Z_0 factors. The medium is isotropic and nonmagnetic, with no explicit source current:

$$\mathbf{D} = \varepsilon\mathbf{E}, \qquad \mathbf{B} = \mathbf{H}, \qquad \varepsilon = 1 + \chi.$$

Here $\chi \equiv \varepsilon - 1$ denotes the dielectric contrast, preserving our X-ray susceptibility notation. The time dependence and vacuum wavenumber are

$$e^{-\mathrm{i}\omega t}, \qquad k_0 = \frac{\omega}{c} = \frac{2\pi}{\lambda}.$$

Curl equations

$$\nabla \times \mathbf{E}(\mathbf{r}) = \mathrm{i}k_0\mathbf{H}(\mathbf{r}). \tag{1}$$

$$\nabla \times \mathbf{H}(\mathbf{r}) = -\mathrm{i}k_0[1 + \chi(\mathbf{r})]\mathbf{E}(\mathbf{r}). \tag{2}$$

Field ansatz in one layer

The structure is periodic along x with period D, invariant along y, and uniform along z within each layer. Depth z increases into the sample.

$$\mathbf{E}(x, y, z) = \sum_h \sum_n e^{\mathrm{i}(k_{x,h}x + k_y y)} \left[T_n \mathbf{E}^+_{hn} e^{\mathrm{i}k_{z,n}z} + R_n \mathbf{E}^-_{hn} e^{-\mathrm{i}k_{z,n}z}\right]. \tag{3}$$

$$\mathbf{H}(x, y, z) = \sum_h \sum_n e^{\mathrm{i}(k_{x,h}x + k_y y)} \left[T_n \mathbf{H}^+_{hn} e^{\mathrm{i}k_{z,n}z} + R_n \mathbf{H}^-_{hn} e^{-\mathrm{i}k_{z,n}z}\right]. \tag{4}$$

$$k_{x,h} = k_x + \frac{2\pi h}{D}.$$

h: diffraction order; n: mode. $\mathbf{E}^{\pm}_{hn}$ and $\mathbf{H}^{\pm}_{hn}$ are three-component Fourier coefficients, constant within a layer. T_n, R_n: forward/backward modal amplitudes, with a common local reference plane $z = 0$. Electric and magnetic fields share these amplitudes; their mode coefficients are related by Maxwell's equations.

2 One forward mode

Fix mode n and omit its overall amplitude T_n. Retain all diffraction orders h. In one layer, define

$$\mathbf{k}_{hn} = (k_{x,h}, k_y, k_{z,n}), \qquad \mathbf{r} = (x, y, z). \tag{5}$$

$$\mathbf{E}^{(n+)}(\mathbf{r}) = \sum_h \mathbf{E}^+_{hn} e^{\mathrm{i}\mathbf{k}_{hn}\cdot\mathbf{r}}, \tag{6}$$

$$\mathbf{H}^{(n+)}(\mathbf{r}) = \sum_h \mathbf{H}^+_{hn} e^{\mathrm{i}\mathbf{k}_{hn}\cdot\mathbf{r}}. \tag{7}$$

First curl

The coefficients are constant in space; the derivative acts on the exponential:

$$\begin{aligned}\nabla \times \mathbf{E}^{(n+)} &= \sum_h \nabla e^{\mathrm{i}\mathbf{k}_{hn}\cdot\mathbf{r}} \times \mathbf{E}^+_{hn} \\ &= \mathrm{i}\sum_h \left(\mathbf{k}_{hn} \times \mathbf{E}^+_{hn}\right) e^{\mathrm{i}\mathbf{k}_{hn}\cdot\mathbf{r}} = \mathrm{i}k_0 \sum_h \mathbf{H}^+_{hn} e^{\mathrm{i}\mathbf{k}_{hn}\cdot\mathbf{r}}.\end{aligned} \tag{8}$$

Match coefficients of each Fourier harmonic and cancel i:

$$\mathbf{k}_{hn} \times \mathbf{E}^+_{hn} = k_0 \mathbf{H}^+_{hn}. \tag{9}$$

Equivalently, in component matrix form:

$$\begin{pmatrix} 0 & -k_{z,n} & k_y \\ k_{z,n} & 0 & -k_{x,h} \\ -k_y & k_{x,h} & 0 \end{pmatrix} \begin{pmatrix} E^+_{x,hn} \\ E^+_{y,hn} \\ E^+_{z,hn} \end{pmatrix} = k_0 \begin{pmatrix} H^+_{x,hn} \\ H^+_{y,hn} \\ H^+_{z,hn} \end{pmatrix}. \tag{10}$$

Second curl

Expand the susceptibility in this layer:

$$\chi(x) = \sum_\ell \chi_\ell e^{\mathrm{i}G_\ell x}, \qquad G_\ell = \frac{2\pi\ell}{D}. \tag{11}$$

Multiplication by χ mixes orders. Since $k_{x,g} + G_\ell = k_{x,g+\ell}$, collecting order h gives $\ell = h - g$:

$$[1 + \chi(x)]\mathbf{E}^{(n+)}(\mathbf{r}) = \sum_h \left[\mathbf{E}^+_{hn} + \sum_g \chi_{h-g} \mathbf{E}^+_{gn} \right] e^{\mathrm{i}\mathbf{k}_{hn}\cdot\mathbf{r}}. \tag{12}$$

Therefore,

$$\begin{aligned} \nabla \times \mathbf{H}^{(n+)} &= \mathrm{i} \sum_h \left(\mathbf{k}_{hn} \times \mathbf{H}^+_{hn} \right) e^{\mathrm{i}\mathbf{k}_{hn}\cdot\mathbf{r}} \\ &= -\mathrm{i}k_0 \sum_h \left[\mathbf{E}^+_{hn} + \sum_g \chi_{h-g} \mathbf{E}^+_{gn} \right] e^{\mathrm{i}\mathbf{k}_{hn}\cdot\mathbf{r}}. \end{aligned} \tag{13}$$

Match Fourier coefficients and cancel i:

$$\mathbf{k}_{hn} \times \mathbf{H}^+_{hn} = -k_0 \left[\mathbf{E}^+_{hn} + \sum_g \chi_{h-g} \mathbf{E}^+_{gn} \right]. \tag{14}$$

Equivalently, in component matrix form:

$$\begin{pmatrix} 0 & -k_{z,n} & k_y \\ k_{z,n} & 0 & -k_{x,h} \\ -k_y & k_{x,h} & 0 \end{pmatrix} \begin{pmatrix} H^+_{x,hn} \\ H^+_{y,hn} \\ H^+_{z,hn} \end{pmatrix} = -k_0 \left[\begin{pmatrix} E^+_{x,hn} \\ E^+_{y,hn} \\ E^+_{z,hn} \end{pmatrix} + \sum_g \chi_{h-g} \begin{pmatrix} E^+_{x,gn} \\ E^+_{y,gn} \\ E^+_{z,gn} \end{pmatrix} \right]. \tag{15}$$

The curl acts separately on each order; the susceptibility couples different orders.

3 The z components

Take the third row of the first-curl matrix, Eq. (10):

$$k_{x,h} E^+_{y,hn} - k_y E^+_{x,hn} = k_0 H^+_{z,hn}. \tag{16}$$

The third row of the second-curl matrix, Eq. (15), gives

$$k_{x,h} H^+_{y,hn} - k_y H^+_{x,hn} = -k_0 \left[E^+_{z,hn} + \sum_g \chi_{h-g} E^+_{z,gn} \right]. \tag{17}$$

Neither equation contains $k_{z,n}$. These relations link the longitudinal components E^+_z, H^+_z to the transverse components $E^+_x, E^+_y, H^+_x, H^+_y$, allowing the longitudinal components to be eliminated before forming the eigenproblem.

4 The x, y components

Take the first two rows of each curl matrix.

First curl

$$k_y E^+_{z,hn} - k_{z,n} E^+_{y,hn} = k_0 H^+_{x,hn}, \tag{18}$$

$$k_{z,n} E^+_{x,hn} - k_{x,h} E^+_{z,hn} = k_0 H^+_{y,hn}. \tag{19}$$

Second curl

$$k_y H^+_{z,hn} - k_{z,n} H^+_{y,hn} = -k_0 \left[E^+_{x,hn} + \sum_g \chi_{h-g} E^+_{x,gn} \right], \tag{20}$$

$$k_{z,n} H^+_{x,hn} - k_{x,h} H^+_{z,hn} = -k_0 \left[E^+_{y,hn} + \sum_g \chi_{h-g} E^+_{y,gn} \right]. \tag{21}$$

Unlike the z-component equations, these four equations contain $k_{z,n}$.

5 Derivation of the eigenvalue problem

For fixed mode n, collect the diffraction orders into columns:

$$\mathbf{E}^+_{\alpha,n} = (E^+_{\alpha,hn})_h, \qquad \mathbf{H}^+_{\alpha,n} = (H^+_{\alpha,hn})_h, \qquad \alpha = x, y, z. \tag{22}$$

We suppress n on these component columns below, but retain the forward-mode superscript $+$. Define

$$(K_x)_{hg} = k_{x,h} \delta_{hg}, \qquad C_{hg} = \delta_{hg} + \chi_{h-g}. \tag{23}$$

C is the permittivity convolution matrix; I is the identity on diffraction orders. The z-component equations become

$$\mathbf{H}^+_z = \frac{1}{k_0} \left(K_x \mathbf{E}^+_y - k_y \mathbf{E}^+_x \right), \tag{24}$$

$$\mathbf{E}^+_z = \frac{1}{k_0} C^{-1} \left(k_y \mathbf{H}^+_x - K_x \mathbf{H}^+_y \right). \tag{25}$$

5.1 First curl: deriving L

Rearrange Eqs. (18)–(19):

$$k_{z,n} \mathbf{E}^+_x = K_x \mathbf{E}^+_z + k_0 \mathbf{H}^+_y, \tag{26}$$

$$k_{z,n} \mathbf{E}^+_y = k_y \mathbf{E}^+_z - k_0 \mathbf{H}^+_x. \tag{27}$$

Substitute Eq. (25):

$$k_{z,n} \mathbf{E}^+_x = \frac{1}{k_0} K_x C^{-1} \left(k_y \mathbf{H}^+_x - K_x \mathbf{H}^+_y \right) + k_0 \mathbf{H}^+_y, \tag{28}$$

$$k_{z,n} \mathbf{E}^+_y = \frac{k_y}{k_0} C^{-1} \left(k_y \mathbf{H}^+_x - K_x \mathbf{H}^+_y \right) - k_0 \mathbf{H}^+_x. \tag{29}$$

Collect coefficients of $\mathbf{H}^+_x, \mathbf{H}^+_y$:

$$k_{z,n} \begin{pmatrix} \mathbf{E}^+_x \\ \mathbf{E}^+_y \end{pmatrix} = \mathsf{L} \begin{pmatrix} \mathbf{H}^+_x \\ \mathbf{H}^+_y \end{pmatrix}, \tag{30}$$

$$\mathsf{L} = \frac{1}{k_0} \begin{pmatrix} k_y K_x C^{-1} & k_0^2 I - K_x C^{-1} K_x \\ k_y^2 C^{-1} - k_0^2 I & -k_y C^{-1} K_x \end{pmatrix}. \tag{31}$$

Matrix order matters: generally $K_x C^{-1} \neq C^{-1} K_x$. The scalar k_y commutes with every matrix.

5.2 Second curl: deriving M

Rearrange Eqs. (20)–(21):

$$k_{z,n}\mathbf{H}_x^+ = K_x\mathbf{H}_z^+ - k_0 C\mathbf{E}_y^+, \tag{32}$$

$$k_{z,n}\mathbf{H}_y^+ = k_y\mathbf{H}_z^+ + k_0 C\mathbf{E}_x^+. \tag{33}$$

Substitute Eq. (24):

$$k_{z,n}\mathbf{H}_x^+ = \frac{1}{k_0}K_x\left(K_x\mathbf{E}_y^+ - k_y\mathbf{E}_x^+\right) - k_0 C\mathbf{E}_y^+, \tag{34}$$

$$k_{z,n}\mathbf{H}_y^+ = \frac{k_y}{k_0}\left(K_x\mathbf{E}_y^+ - k_y\mathbf{E}_x^+\right) + k_0 C\mathbf{E}_x^+. \tag{35}$$

Collect coefficients of $\mathbf{E}_x^+, \mathbf{E}_y^+$:

$$k_{z,n}\begin{pmatrix}\mathbf{H}_x^+\\ \mathbf{H}_y^+\end{pmatrix} = \mathsf{M}\begin{pmatrix}\mathbf{E}_x^+\\ \mathbf{E}_y^+\end{pmatrix}, \tag{36}$$

$$\mathsf{M} = \frac{1}{k_0}\begin{pmatrix}-k_yK_x & K_x^2 - k_0^2C\\ k_0^2C - k_y^2I & k_yK_x\end{pmatrix}. \tag{37}$$

5.3 Combining the two relations

Define the transverse modal columns

$$\mathcal{E}_{t,n}^+ = \begin{pmatrix}\mathbf{E}_{x,n}^+\\ \mathbf{E}_{y,n}^+\end{pmatrix}, \qquad \mathcal{H}_{t,n}^+ = \begin{pmatrix}\mathbf{H}_{x,n}^+\\ \mathbf{H}_{y,n}^+\end{pmatrix}. \tag{38}$$

The two curl relations are

$$k_{z,n}\mathcal{E}_{t,n}^+ = \mathsf{L}\mathcal{H}_{t,n}^+, \qquad k_{z,n}\mathcal{H}_{t,n}^+ = \mathsf{M}\mathcal{E}_{t,n}^+. \tag{39}$$

Multiply the first by $k_{z,n}$ and substitute the second:

$$k_{z,n}^2\mathcal{E}_{t,n}^+ = \mathsf{L}\left(k_{z,n}\mathcal{H}_{t,n}^+\right) = \mathsf{LM}\mathcal{E}_{t,n}^+. \tag{40}$$

Thus

$$\left(\mathcal{A} - k_{z,n}^2 I\right)\mathcal{E}_{t,n}^+ = 0, \qquad \mathcal{A} = \mathsf{LM}. \tag{41}$$

For N diffraction orders, $\mathcal{A}$ is $2N \times 2N$. Its eigenvalues are $k_{z,n}^2$ and its eigenvectors contain the transverse electric coefficients for all orders.

Finite Fourier implementation. The derivation above uses direct permittivity convolution. In the code, inverse Fourier factorization replaces C in the lower-left block of M by $C_x = ([1/\varepsilon])^{-1}$, where $[1/\varepsilon]_{hg} = (1/\varepsilon)_{h-g}$. The matrix L and the elimination leading to $\mathcal{A} = \mathsf{LM}$ are unchanged.

5.4 Forward and backward modes

The eigenproblem determines $k_{z,n}^2$, so each eigenvalue supplies two propagation constants:

$$k_{z,n}^{(+)} = +k_{z,n}, \qquad k_{z,n}^{(-)} = -k_{z,n}, \qquad \left(k_{z,n}^{(+)}\right)^2 = \left(k_{z,n}^{(-)}\right)^2. \tag{42}$$

Their depth factors are $e^{+\mathrm{i}k_{z,n}z}$ and $e^{-\mathrm{i}k_{z,n}z}$. For passive media, select the forward root with $\operatorname{Im} k_{z,n} \geq 0$; for propagating lossless modes, take $k_{z,n} > 0$.

Final eigenvalue problem

$$\boxed{\left(\mathcal{A} - k_{z,n}^2 I\right)\mathcal{E}_{t,n}^{\pm} = 0, \qquad \mathcal{A} = \mathsf{LM}.} \tag{43}$$

The same transverse electric eigenvector can be used for the paired branches:

$$\mathcal{E}_{t,n}^- = \mathcal{E}_{t,n}^+. \tag{44}$$

This fixes a common normalization for the pair; the physical amplitudes T_n and R_n remain independent. No second eigenproblem is required.

Recovering the magnetic modes

Use the electric eigenvector and its signed propagation constant:

$$\mathcal{H}_{t,n}^{\pm} = \frac{1}{k_{z,n}^{(\pm)}} \mathsf{M} \mathcal{E}_{t,n}^{\pm}, \qquad k_{z,n} \neq 0. \tag{45}$$

Consequently,

$$\mathcal{H}_{t,n}^{-} = -\mathcal{H}_{t,n}^{+}. \tag{46}$$

The values $k_{z,n}$ alone are insufficient: the corresponding electric eigenvectors are also needed.

Longitudinal components and sign relations

Recover the remaining component columns from the z equations:

$$\mathbf{E}_{z,n}^{\pm} = \frac{1}{k_0} C^{-1} \left(k_y \mathbf{H}_{x,n}^{\pm} - K_x \mathbf{H}_{y,n}^{\pm} \right), \tag{47}$$

$$\mathbf{H}_{z,n}^{\pm} = \frac{1}{k_0} \left(K_x \mathbf{E}_{y,n}^{\pm} - k_y \mathbf{E}_{x,n}^{\pm} \right). \tag{48}$$

Thus the full three-component coefficients obey, for every order h,

$$\mathbf{E}_{hn}^{-} = \begin{pmatrix} E_{x,hn}^{+} \\ E_{y,hn}^{+} \\ -E_{z,hn}^{+} \end{pmatrix}, \qquad \mathbf{H}_{hn}^{-} = \begin{pmatrix} -H_{x,hn}^{+} \\ -H_{y,hn}^{+} \\ H_{z,hn}^{+} \end{pmatrix}. \tag{49}$$

Reversing propagation leaves the tangential electric and normal magnetic components unchanged; it reverses the normal electric and tangential magnetic components.

6 Boundary conditions

Consider a sharp horizontal interface between layers j and $j+1$, with no surface currents or free surface charge. The tangential fields are continuous:

$$E_x^{(j)} = E_x^{(j+1)}, \qquad E_y^{(j)} = E_y^{(j+1)}, \qquad H_x^{(j)} = H_x^{(j+1)}, \qquad H_y^{(j)} = H_y^{(j+1)}. \tag{50}$$

These equalities hold at the same interface plane and provide four equations per diffraction order.

Modal matrices

Assemble the forward transverse eigenvectors into columns:

$$\mathcal{E}_t^{(j)} = \left(\mathcal{E}_{t,1}^{(j)+}, \ldots, \mathcal{E}_{t,2N}^{(j)+} \right), \qquad \mathcal{H}_t^{(j)} = \left(\mathcal{H}_{t,1}^{(j)+}, \ldots, \mathcal{H}_{t,2N}^{(j)+} \right). \tag{51}$$

Using $\mathcal{E}_t^{-} = \mathcal{E}_t^{+}$ and $\mathcal{H}_t^{-} = -\mathcal{H}_t^{+}$, the total tangential field at depth z is

$$\begin{pmatrix} \mathbf{E}_x^{(j)}(z) \\ \mathbf{E}_y^{(j)}(z) \\ \mathbf{H}_x^{(j)}(z) \\ \mathbf{H}_y^{(j)}(z) \end{pmatrix} = \mathcal{P}_j \begin{pmatrix} \mathbf{T}_j(z) \\ \mathbf{R}_j(z) \end{pmatrix}, \qquad \mathcal{P}_j = \begin{pmatrix} \mathcal{E}_t^{(j)} & \mathcal{E}_t^{(j)} \\ \mathcal{H}_t^{(j)} & -\mathcal{H}_t^{(j)} \end{pmatrix}. \tag{52}$$

Here $\mathbf{T}_j(z), \mathbf{R}_j(z)$ are modal amplitudes evaluated at z, including their propagation phases. $\mathcal{P}_j$ is $4N \times 4N$.

Matching at an interface

At $z = z_b$, continuity becomes

$$\mathcal{P}_j \begin{pmatrix} \mathbf{T}_j(z_b) \\ \mathbf{R}_j(z_b) \end{pmatrix} = \mathcal{P}_{j+1} \begin{pmatrix} \mathbf{T}_{j+1}(z_b) \\ \mathbf{R}_{j+1}(z_b) \end{pmatrix}. \tag{53}$$

Thus, in the familiar interface-transfer notation,

$$\boxed{\begin{pmatrix} \mathbf{T}_j(z_b) \\ \mathbf{R}_j(z_b) \end{pmatrix} = \mathcal{V}_{j,j+1} \begin{pmatrix} \mathbf{T}_{j+1}(z_b) \\ \mathbf{R}_{j+1}(z_b) \end{pmatrix}, \qquad \mathcal{V}_{j,j+1} = \mathcal{P}_j^{-1} \mathcal{P}_{j+1}.} \tag{54}$$

Propagation between interfaces

Let z_j be the top of layer j, with thickness d_j and $z_{j+1} = z_j + d_j$. Define

$$\mathcal{Q}_j(d_j) = \begin{pmatrix} \mathrm{diag}(e^{+\mathrm{i}k_{z,n}^{(j)} d_j}) & 0 \\ 0 & \mathrm{diag}(e^{-\mathrm{i}k_{z,n}^{(j)} d_j}) \end{pmatrix}. \tag{55}$$

For amplitudes referenced to the respective layer tops:

$$\mathcal{P}_j \mathcal{Q}_j(d_j) \begin{pmatrix} \mathbf{T}_j(z_j) \\ \mathbf{R}_j(z_j) \end{pmatrix} = \mathcal{P}_{j+1} \begin{pmatrix} \mathbf{T}_{j+1}(z_{j+1}) \\ \mathbf{R}_{j+1}(z_{j+1}) \end{pmatrix}. \tag{56}$$

The signs follow our positive-depth convention. For numerical stability, the code anchors forward amplitudes at layer tops and backward amplitudes at bottoms, so traversal uses only $e^{+\mathrm{i}k_{z,n}^{(j)} d_j}$.

Closing the system

$$\mathbf{T}_{\text{ambient}} = \mathbf{T}_{\text{incident}}, \qquad \mathbf{R}_{\text{substrate}} = 0. \tag{57}$$

The chosen s or p incident electric field fixes $\mathbf{T}_{\text{incident}}$ in the ambient modal basis. No wave is incident from the substrate. Solving the interface equations determines all remaining $\mathbf{T}_j, \mathbf{R}_j$.

6.1 Incident s and p polarization

The interface conditions are identical for s and p. For the EUV and X-ray geometry considered here, we assume vacuum ambient: $n_{\mathrm{a}} = 1$, $\varepsilon_{\mathrm{a}} = 1$. Polarization enters through the prescribed incident field.

Geometry and polarization vectors

Let α be the grazing angle above the surface and ϕ the azimuth measured from $+y$ towards $+x$. Then

$$\widehat{\mathbf{k}}_{\text{inc}} = \begin{pmatrix} \cos\alpha \sin\phi \\ \cos\alpha \cos\phi \\ \sin\alpha \end{pmatrix}, \qquad \mathbf{k}_{\text{inc}} = k_0 \widehat{\mathbf{k}}_{\text{inc}}. \tag{58}$$

Choose unit electric polarization vectors

$$\mathbf{s} = \begin{pmatrix} \cos\phi \\ -\sin\phi \\ 0 \end{pmatrix}, \qquad \mathbf{p} = \widehat{\mathbf{k}}_{\text{inc}} \times \mathbf{s} = \begin{pmatrix} \sin\alpha \sin\phi \\ \sin\alpha \cos\phi \\ -\cos\alpha \end{pmatrix}. \tag{59}$$

In Gaussian units, $\mathbf{H}_{\text{inc}} = \widehat{\mathbf{k}}_{\text{inc}} \times \mathbf{E}_{\text{inc}}$:

Polarization	$\mathbf{E}_{\text{inc}}$	$\mathbf{H}_{\text{inc}}$
s	$\mathbf{s}$	$\mathbf{p}$
p	$\mathbf{p}$	$-\mathbf{s}$

Both choices have $|\mathbf{E}_{\text{inc}}| = 1$.

Selecting the incident diffraction order

For $N = 2M + 1$ orders $h = -M, \ldots, M$, define the Kronecker-delta column

$$d_h = \delta_{h0}, \qquad \mathbf{d} = (\ldots, 0, 0, 1, 0, 0, \ldots)^{\mathsf{T}} \in \mathbb{C}^N. \tag{60}$$

The central entry selects $h = 0$; all other incident orders vanish. In the scalar formulation this column alone specifies a unit incident wave. Here there are $2N$ vector modes and $\mathbf{T}_{\text{inc}} \in \mathbb{C}^{2N}$.

Ambient basis used in the code

For each order, choose one mode with unit tangential E_x and one with unit tangential E_y. Arrange the x modes first and the y modes second, so $\mathcal{E}_t^{(\mathrm{a})} = I_{2N}$. Then

$$\mathcal{E}_t^{(\mathrm{a})} \mathbf{T}_{\text{inc}} = \begin{pmatrix} E_{\text{inc},x} \mathbf{d} \\ E_{\text{inc},y} \mathbf{d} \end{pmatrix}. \tag{61}$$

Therefore the final incident-amplitude columns are

$$\boxed{\mathbf{T}^{(s)}_{\text{inc}} = \begin{pmatrix} \cos\phi\,\mathbf{d} \\ -\sin\phi\,\mathbf{d} \end{pmatrix}} \qquad \boxed{\mathbf{T}^{(p)}_{\text{inc}} = \begin{pmatrix} \sin\alpha\sin\phi\,\mathbf{d} \\ \sin\alpha\cos\phi\,\mathbf{d} \end{pmatrix}}. \tag{62}$$

The x/y block arrangement refers to this ambient basis; grating eigenmodes generally mix both transverse components.

The longitudinal electric component follows from $\mathbf{k}_{\text{inc}} \cdot \mathbf{E}_{\text{inc}} = 0$: it is 0 for s and $-\cos\alpha$ for p. The magnetic field follows from the first curl, reproducing the table above. Neither requires an independent incident-amplitude column. The substrate condition remains $\mathbf{R}_{\text{substrate}} = 0$.

7 Semi-analytic modal integration

Consider one layer of thickness d, with local $z = 0$ at its top. Let $p(x)$ be the fluorescent-atom density weight, independent of depth within this layer. Without fluorescence escape attenuation, the relative yield per unit length along y is

$$Y_j \propto \int_0^d \mathrm{d}z \int_{\text{period}} \mathrm{d}x\, p(x) \sum_{\alpha=x,y,z} |E_\alpha(x,z)|^2. \tag{63}$$

Both incident polarizations use total electric intensity. All amplitudes in the following derivation are referenced to the same layer top.

Combining forward and backward modes

For each Cartesian component, assemble the order-by-mode matrices

$$E^{\pm}_\alpha = (E^{\pm}_{\alpha,hn})_{h,n}, \qquad \mathbb{E}_\alpha = \begin{pmatrix} E^+_\alpha & E^-_\alpha \end{pmatrix}, \qquad \alpha = x, y, z. \tag{64}$$

Define a combined amplitude column and signed propagation constants:

$$\mathbf{c} = \begin{pmatrix} \mathbf{T} \\ \mathbf{R} \end{pmatrix}, \qquad \boldsymbol{\kappa} = (k_{z,1}, \ldots, k_{z,2N}, -k_{z,1}, \ldots, -k_{z,2N}). \tag{65}$$

The combined indices $a, b = 1, \ldots, 4N$ run over both propagation directions. At $y = 0$, the field is

$$E_\alpha(x,z) = \sum_h \sum_a (\mathbb{E}_\alpha)_{ha}\, c_a\, e^{\mathrm{i}k_{x,h}x} e^{\mathrm{i}\kappa_a z}. \tag{66}$$

The component matrices retain the sign relations $E^-_x = E^+_x$, $E^-_y = E^+_y$, $E^-_z = -E^+_z$.

Lateral integration

$$\Phi_{hg} = \int_{\text{period}} p(x) e^{\mathrm{i}(k_{x,h} - k_{x,g})x}\, \mathrm{d}x. \tag{67}$$

The order of indices fixes the sign of the exponential: row h minus column g.

Depth integration

$$U(q) = \int_0^d e^{\mathrm{i}qz}\, \mathrm{d}z = \frac{e^{\mathrm{i}qd} - 1}{\mathrm{i}q}, \qquad U(0) = d. \tag{68}$$

$$\Psi_{ab} = c_a c_b^*\, U(\kappa_a - \kappa_b^*). \tag{69}$$

The four blocks of Ψ contain the $T_n T_m^*$, $T_n R_m^*$, $R_n T_m^*$, and $R_n R_m^*$ contributions, including all mode-pair interference.

Final matrix expression

$$\mathcal{F} = \Phi \odot \left(\mathbb{E}_x \Psi \mathbb{E}_x^\dagger + \mathbb{E}_y \Psi \mathbb{E}_y^\dagger + \mathbb{E}_z \Psi \mathbb{E}_z^\dagger \right), \tag{70}$$

$$Y_j \propto \sum_{h,g} \mathcal{F}_{hg}, \qquad Y = \sum_j Y_j. \tag{71}$$

Here $\odot$ denotes elementwise multiplication and $\dagger$ conjugate transpose. For N orders, $\Phi, \mathcal{F}$ are $N \times N$, Ψ is $4N \times 4N$, and each $\mathbb{E}_\alpha$ is $N \times 4N$. Sum all entries of $\mathcal{F}$, not only its diagonal.

Relation to the implementation. Use coefficients of the physical electric field. With inverse Fourier factorization, the code evaluates this expression separately in each nominal material region, reconstructing E_x from D_x/ε, and sums the region contributions. For numerical stability, it uses the equivalent depth integral with forward amplitudes referenced to the top and backward amplitudes to the bottom, rather than explicitly converting them to a common reference plane.

8 Line-edge roughness

Sidewall roughness is represented by Gaussian damping of each box's lateral Fourier coefficients. Let $\sigma_{\mathrm{LE},b}$ be the width in nm for box b, and $G_m = 2\pi m/D$. The Debye–Waller factor is

$$W_{m,b} = \exp\left(-\frac{G_m^2 \sigma_{\mathrm{LE},b}^2}{2}\right). \tag{72}$$

For either $f = \varepsilon$ or $f = 1/\varepsilon$, the sharp-box contrast is

$$\Delta f_m^{(b)} = \frac{f_b - f_{\mathrm{bg}}}{D} \int_{x_b^-}^{x_b^+} e^{-\mathrm{i}G_m x}\,\mathrm{d}x. \tag{73}$$

Apply the factor before assembling the convolution matrices:

$$\widehat{f}_m = f_{\mathrm{bg}}\delta_{m0} + \sum_b \Delta f_m^{(b)} W_{m,b}. \tag{74}$$

Thus

$$C_{hg} = \widehat{\varepsilon}_{h-g}, \qquad B_{hg} = \widehat{(1/\varepsilon)}_{h-g}, \qquad C_x = B^{-1}. \tag{75}$$

The same factor damps permittivity and reciprocal-permittivity contrasts. Since $W_{0,b} = 1$, the zero harmonic is unchanged. Setting $\sigma_{\mathrm{LE},b} = 0$ recovers the sharp box. In the API this is `Box.sigma`, or the `sigma` argument of `Layer.add_box`, defaulting to 0 nm.

This is effective Fourier averaging within a model invariant along y; it does not resolve random edge variations along the lines or diffuse scattering. Separately averaged ε and $1/\varepsilon$ are not generally pointwise reciprocals. Field reconstruction and fluorescence retain the nominal material regions; the fluorescent-density overlaps Φ are not damped. Consequently nominal-volume electric loss need not equal the loss inferred from the averaged constitutive operators.

9 Interfacial roughness

For the horizontal interface between layers j and $j+1$, let σ_j be the RMS height roughness in nm. Start with the sharp-interface matrix

$$\mathcal{V}_{j,j+1} = \mathcal{P}_j^{-1}\mathcal{P}_{j+1}. \tag{76}$$

In each layer, collect signed vertical wavevectors in the same order as the forward/backward amplitudes:

$$\boldsymbol{\kappa}^{(j)} = \left(k_{z,1}^{(j)}, \ldots, k_{z,2N}^{(j)}, -k_{z,1}^{(j)}, \ldots, -k_{z,2N}^{(j)}\right). \tag{77}$$

Gaussian interface averaging gives the modal factor

$$F_{ab}^{(j)} = \exp\left[-\frac{\sigma_j^2}{2}\left(\kappa_a^{(j)} - \kappa_b^{(j+1)}\right)^2\right], \qquad \mathcal{V}_{j,j+1}^{(\sigma)} = \mathcal{V}_{j,j+1} \odot F^{(j)}. \tag{78}$$

$F^{(j)}$ is generally a dense $4N \times 4N$ matrix. The exponential and the multiplication are elementwise, not matrix operations. Use the complex signed wavevectors: there is no conjugation or absolute-value square here. At $\sigma_j = 0$, every entry of $F^{(j)}$ is one.

The corrected matching equation at the nominal interface plane z_b is

$$\begin{pmatrix}\mathbf{T}_j(z_b)\\ \mathbf{R}_j(z_b)\end{pmatrix} = \mathcal{V}_{j,j+1}^{(\sigma)} \begin{pmatrix}\mathbf{T}_{j+1}(z_b)\\ \mathbf{R}_{j+1}(z_b)\end{pmatrix}. \tag{79}$$

This modifies interface matching; the layer propagation matrix $\mathcal{Q}_j$ is unchanged.

Homogeneous-interface limit

The prescription recovers the complex Névot–Croce factors for tangential electric reflection and transmission amplitudes, for both s and p:

$$r_\sigma = r_0 \exp(-2k_{z,j}k_{z,j+1}\sigma_j^2), \qquad t_\sigma = t_0 \exp\left[\frac{(k_{z,j} - k_{z,j+1})^2\sigma_j^2}{2}\right]. \tag{80}$$

Here r_0, t_0 are the sharp-interface Fresnel amplitudes; see Haase et al. (2016), Eq. 3. The application to coupled grating modes is an effective modal extension.

In the API, `Layer.sigma` belongs to the layer's top interface, and `Structure.substrate_sigma` to the substrate top. Both default to 0 nm and are independent of `Box.sigma`. Artificial slice boundaries should remain sharp.

The corrected equation describes effective fields at nominal interfaces. Diffuse fields and roughness correlations are not modeled. Fluorescence retains the nominal material regions. Exact flux conservation is not enforced: $1 - R - T$ is not pure absorption or a calculated diffuse loss, and can be negative. Large roughness or strongly evanescent modes may invalidate this approximation; overflowing interface factors are reported rather than clipped.